

ZEYU (STEVEN) ZHANG

Ph.D. Student in Statistics and Data Science, Northwestern University

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RESEARCH INTERESTS

Trustworthy and reliable large language models. My current research centers on **temporal knowledge leakage in LLM backtesting**: detecting and attributing contamination in model reasoning with interpretable, claim-level metrics, and eliminating it through inference-time supervision and reinforcement-learning post-training. Related interests include black-box LLM interpretability and auditing, RL-based fine-tuning (GRPO, DPO, RLHF), and factual memorization in LLMs.

EDUCATION

Northwestern University

Evanston, IL, USA

Sept 2023 – Present

Ph.D. in Statistics and Data Science GPA: 3.95/4.00

- Advisor: Prof. Bradley C. Stadie. Expected graduation: June 2028.

University of Science and Technology of China (USTC)

Hefei, China

Sept 2019 – Jun 2023

B.Eng. in Electronic Information Engineering; Minor in Artificial Intelligence GPA: 3.93/4.30

- Rank 5/213 in the School of Information Science and Technology.
- China National Scholarship, Ministry of Education of the PRC (top 1%).
- Wang Xiaomo Talent Program in Cyber Science and Technology; Talent Program in Artificial Intelligence.

PUBLICATIONS & PREPRINTS

Conference Proceedings

[C1] LAMP: Extracting Locally Linear Decision Surfaces from LLM World Models

Ryan Chen, Youngmin Ko, Zeyu Zhang, Catherine Cho, Sunny Chung, Mauro Giuffr , Dennis L. Shung, Bradley C. Stadie
International Conference on Artificial Intelligence and Statistics (AISTATS), 2026. **Spotlight** [arXiv] [OpenReview]

[C2] Unified Off-Policy Learning to Rank: a Reinforcement Learning Perspective

Zeyu Zhang, Yi Su, Hui Yuan, Yiran Wu, Rishab Balasubramanian, Qingyun Wu, Huazheng Wang, Mengdi Wang
Advances in Neural Information Processing Systems (NeurIPS), 2023. [arXiv] [OpenReview] [Code]

Preprints (Under Review)

[P1] Temporal Leakage in LLM Backtesting: Measurement, Validation, and Adjusted Scores

Zeyu Zhang, Bradley C. Stadie
In submission to TMLR. [Code]

[P2] TEMPO: Temporal Enforcement via Mode-Separated Policy Optimization for Trustworthy LLM Backtesting

Zeyu Zhang, Bradley C. Stadie
Under review at NeurIPS 2026. [arXiv:2605.18843] [Code]

[P3] All Leaks Count, Some Count More: Interpretable Temporal Contamination Detection and Mitigation in LLM Backtesting

Zeyu Zhang, Ryan Chen, Bradley C. Stadie
Under review at EMNLP 2026. [arXiv:2602.17234] [Code]

RESEARCH EXPERIENCE

Trustworthy LLM Backtesting: Leakage Theory, Detection & Enforcement

Northwestern University

Sept 2025 – Present

Advisor: Prof. Bradley C. Stadie → papers [P1], [P2], [P3]

- Formalized **temporal knowledge leakage** in LLM backtesting: models exploit post-cutoff pre-training knowledge when evaluated on historical prediction tasks, inflating reported accuracy and invalidating evaluation.
- Built the statistical foundation of the program: showed the standard pre/post contamination check is uninformative (recency alone reproduces the leakage signature; four flagships fail it on questions they cannot have memorized), proved the inflation is *non-identifiable* from passive scores, and showed one external reference restores a **leakage-adjusted score** with confidence intervals—validated by planting leakage in twin models and deployed to audit frontier systems.

- Introduced **Shapley-DCLR**, a claim-level metric that decomposes prediction rationales into atomic claims and applies Shapley values to quantify what fraction of *decision-critical* reasoning is contaminated—making contamination measurable and interpretable.
- Designed **TimeSPEC**, an inference-time architecture that interleaves temporally filtered retrieval with claim-level supervision, grounding predictions entirely in pre-cutoff evidence without any retraining; ablations across three LLMs show retrieval and supervision are jointly necessary.
- Developed **TEMPO**, a GRPO-based post-training method with a two-mode reward that drives leaked claims to zero as a hard prerequisite before optimizing task performance; proved the training procedure monotonically decreases leakage and converges to the leak-free optimum.
- Reduced leakage from 2–13% to 0.6–3.7% across three prediction tasks and two models, while *improving* task performance by 6–13% where strong pre-cutoff signals exist.

LAMP: Black-Box Interpretability via Locally Linear Decision Surfaces

Northwestern University

Jan 2024 – Jan 2025

Advisor: Prof. Bradley C. Stadie → paper [C1], AISTATS 2026 Spotlight

- Co-developed the **Linear Attribution Mapping Probe (LAMP)**, which treats an LLM's self-reported explanations as a coordinate system and fits locally linear surrogates to test whether stated factors actually steer the model's predictions.
- Designed perturbation-based probing that extracts decision surfaces *without* gradients, logits, or internal activations—enabling practical auditing of proprietary LLMs.
- Ran experiments across sentiment analysis, controversial-topic detection, and safety-prompt auditing; extracted surfaces align with human judgments and, on a clinical case-file dataset, with expert assessments.

Factual Memorization and Learning in Large Language Models

Northwestern University

Jan 2024 – Jan 2025

Advisor: Prof. Bradley C. Stadie (Qualifying Exam research)

- Analyzed LLM memorization under SFT and DPO fine-tuning, reproducing and extending Stanford's FINETUNE BENCH; evaluated robustness to rephrasings and temporal shifts.
- Proposed a formal distinction between **passive memorization** (exposure) and **positive memorization** (direct QA supervision), showing the latter is markedly more efficient.
- Showed injected future-dated facts do not generalize beyond training, and that a lightweight system prompt enforcing temporal consistency mitigates the resulting overfitting.

CUOLR: Unified Off-Policy Learning to Rank

Princeton University (remote)

May 2022 – Dec 2023

Mentors: Prof. Mengdi Wang, Prof. Huazheng Wang → paper [C2], NeurIPS 2023

- Unified ranking under general stochastic click models as a **Markov Decision Process**, enabling principled application of offline RL to off-policy learning to rank.
- Proposed **CUOLR**, a click-model-agnostic algorithm requiring no explicit debiasing or prior knowledge of the click process; instantiated with DQN, SAC, BCQ, and CQL.
- Achieved state-of-the-art performance on large-scale real-world benchmarks with consistent robustness across heterogeneous click models.

TEACHING EXPERIENCE

Graduate Teaching Assistant, Department of Statistics and Data Science, Northwestern University.

STAT 303-3	Data Science 3 with Python	(Prof. Emre Besler)	Spring 2025, Spring 2026
STAT 303-2	Data Science 2 with Python	(Prof. Emre Besler)	Winter 2025, Winter 2026
STAT 348	Applied Multivariate Analysis	(Prof. Thomas Severini)	Fall 2025
STAT 210	Introduction to Probability and Statistics	(Prof. Maxim Sinitsyn)	Fall 2024

HONORS & AWARDS

- **Spotlight Presentation**, AISTATS 2026 (LAMP). 2026
- **China National Scholarship**, Ministry of Education of the PRC (top 1%). 2020
- **Outstanding Student Scholarship** (Class A, top 1%), USTC. 2020
- National Undergraduate Electronic Design Contest: 2nd Prize (national), 1st in Anhui Province. 2021
- Scholarship for Talent Program in Basic Disciplines (Class A, top 3%), USTC. 2020

SKILLS

- **Programming:** Python (PyTorch, Hugging Face Transformers), R, C/C++, Bash, Git, $\LaTeX$.
- **ML / LLM:** RL post-training (GRPO, DPO, RLHF), LLM evaluation and benchmarking, retrieval-augmented pipelines, offline reinforcement learning, API-based LLM systems (OpenAI platform).